

Final Project: Evaluating the Intensity of the Urban Heat Island in San Salvador, El Salvador

EAS 508: Python for Earth Science Applications

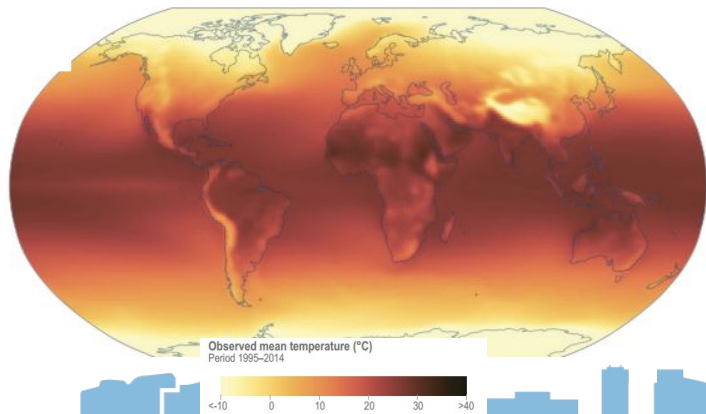
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1. Background

Heatwaves and the Tropics

- ❖ Extreme heat is the leading cause of death globally related to weather and climate (WMO, 2024).
- ❖ People living in the tropics are highly exposed to extreme heat events - high temperatures and high humidity (Jamei et al., 2020).
 - 40% of the world's population resides in the tropics (Wilkinson, 2014).
 - For instance, between 50 and 80% of Central America's population live in exposed, densely populated urban areas (UN-Habitat, 2023).
 - The frequency of long heatwaves in the tropics is also increasing faster than in other parts of the globe (Martinez-Villalobos et al., 2025).



Observed mean temperature between 1985 - 2014.
Image credit: IPCC Sixth Assessment Report, 2022)

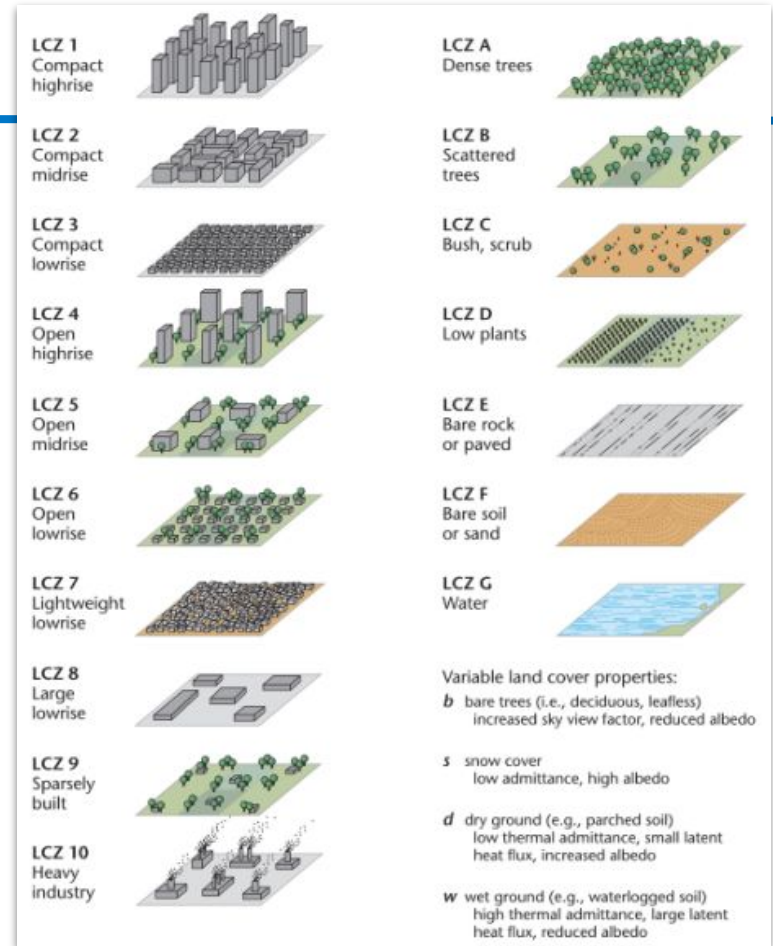


50 to 80% of Central America's population lives in urban areas. Image credit: UN-Habitat (2023)

1. Background

Local Climate Zones - LCZs (categorical)

“Regions of uniform surface cover, structure, material, and human activity that span hundreds of meters to several kilometers horizontally” (Stewart & Oke 2012). “Each LCZ has a specific range of climate-relevant physical parameters. Therefore, the LCZ system establishes a fundamental research framework for urban climate studies and standardizes the worldwide exchange of urban climate knowledge” (Huang et al., 2023).



Classification of Local Climate Zones (LCZ) according to their perceived ability to modify local climate. Image credit: Oke et al. (2012)

1. Background

Urban Heat Island (UHI) effect

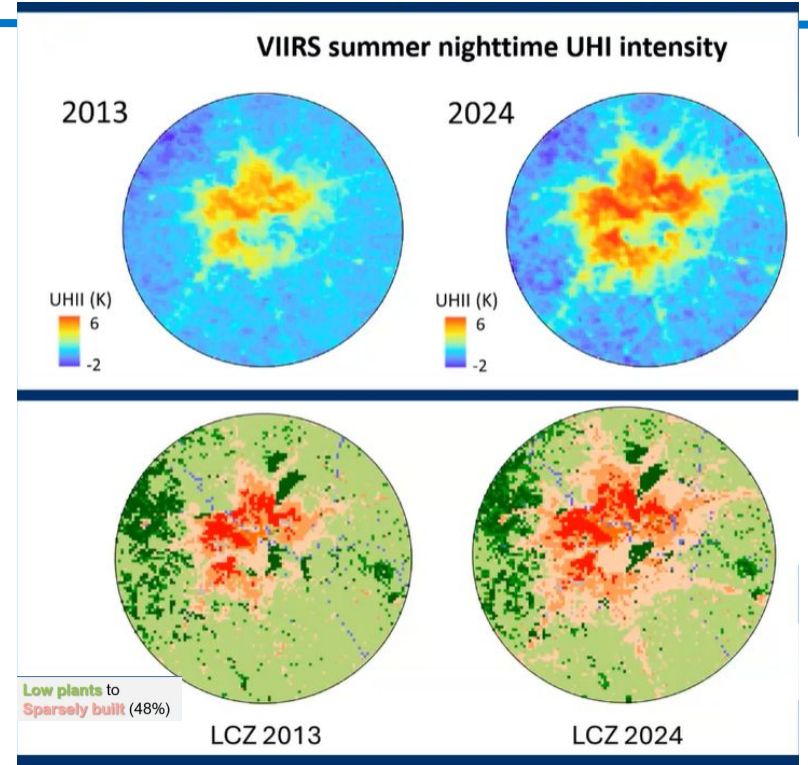
Factors associated with urbanization such as building surface materials (roof types), surface albedos, canopy coverage, and sky view can alter the temperature and can amplify anthropogenic warming in urban areas, triggering the UHI effect (Masson et al. 2020).

Urban Heat Island Intensity (UHII) index

“The UHI intensity (UHII) is the main way to quantify the strength of the UHI signal, reflecting the average temperature difference between the urban area and its background reference area (BRA)” ([Yang et al., 2024](#)).

$$UHII = \Delta T_{u-r} = T_u - T_r$$

The urban heat island intensity (UHII) is estimated by comparing the average and maximum temperature between urban (T_u) and rural area (T_r) (Oke, 1973)



Illustrative example of urban expansion and the urban heat island effect.
Image credit: Yoo et al. (2025) - presented by G. Hulley (NASA JPL)

2. Objectives

This research assesses the spatial patterns of LST across the different LCZs in the city of San Salvador, El Salvador, comparing and analyzing the differences in land surface temperature between the most compact, impervious, and high-rise urban areas (LCZs 1-3), with more vegetated and sparse rural areas (LCZs 6, 9).

This study's main objectives include:

- Identifying the locations of areas most exposed to urban heat island (UHI) effect during 2025,
- Evaluating LST differences between compact, impervious, and high rise urban areas, and more vegetated and sparse rural areas.

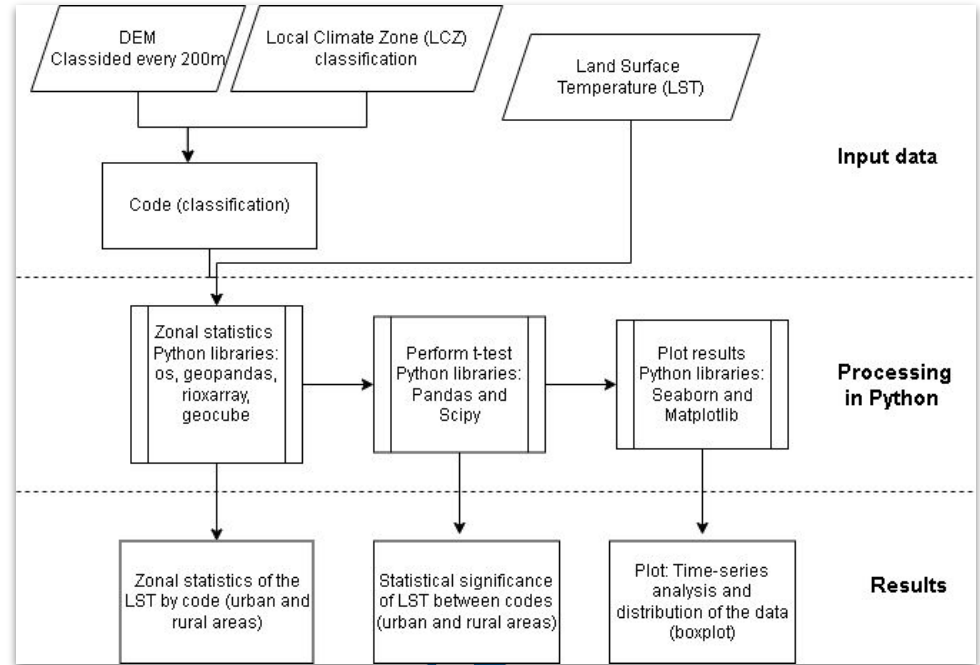
This research aims to provide municipal decision-makers from San Salvador with data that can be used to develop strategies to mitigate UHI effects.



2. Methods

To identify areas most exposed to heat, I used the LCZ and elevation classification to develop a code (classes) that discriminated urban (LCZs 1, 2, 3) from rural (LCZs 6, 9) areas. Using this classification and the LST, I used zonal statistics, t-test analysis and plotted the results using Python libraries as shown in the workflow on the right.

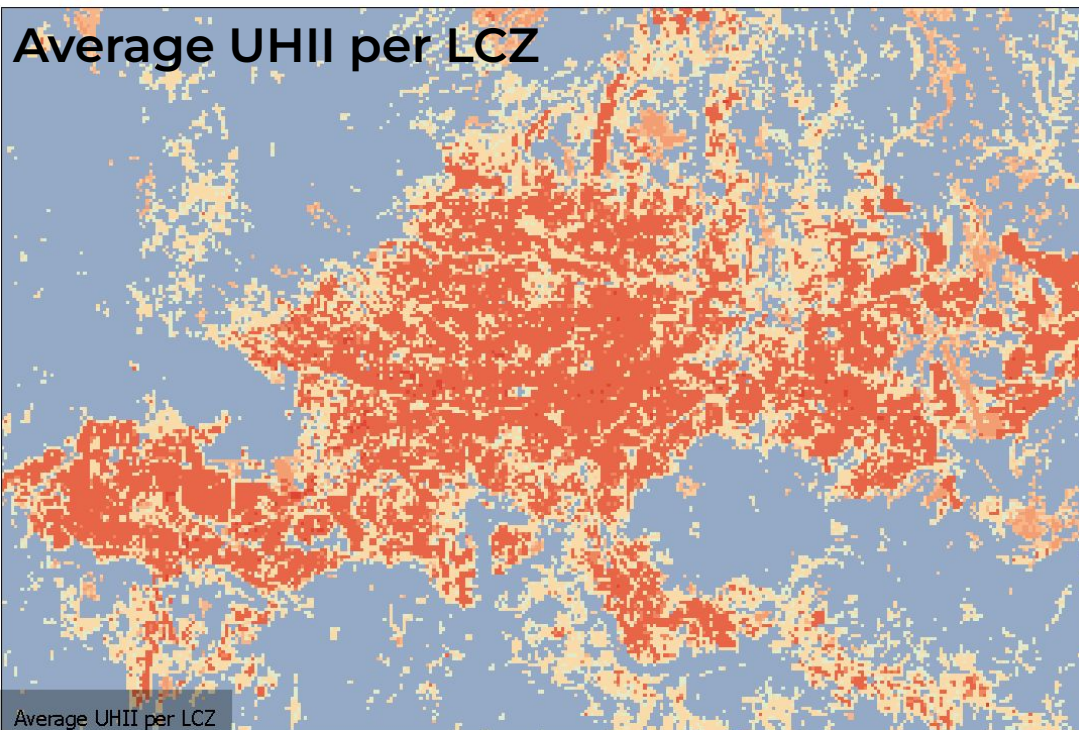
The Python scripts can be found in [GitHub](#).



Workflow of this project

3. Results

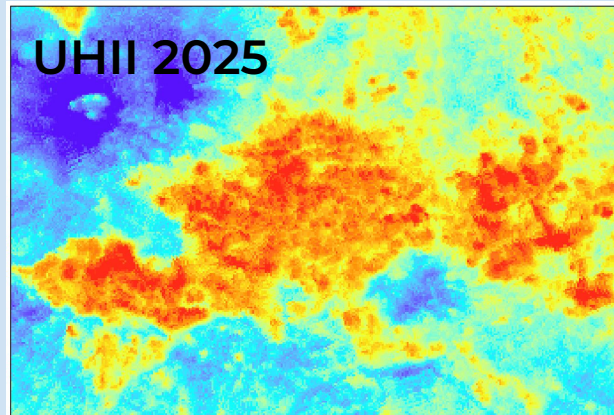
UHI Intensity (UHII)



Average UHII per LCZ

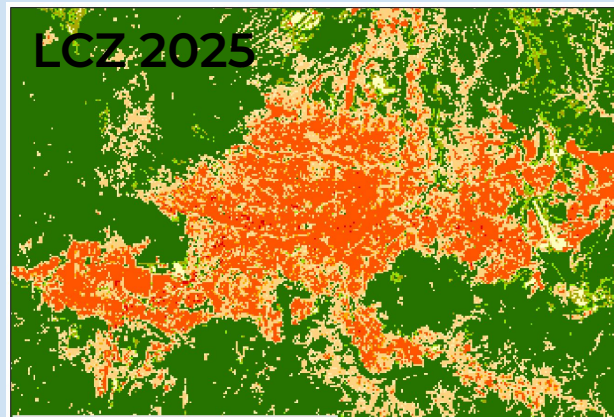
(°C)
7
-4

Analysis based on LST, Planet imagery, and Open Building v. 2.5 data.



UHII (°C)
17
-12

Analysis based on LST data.

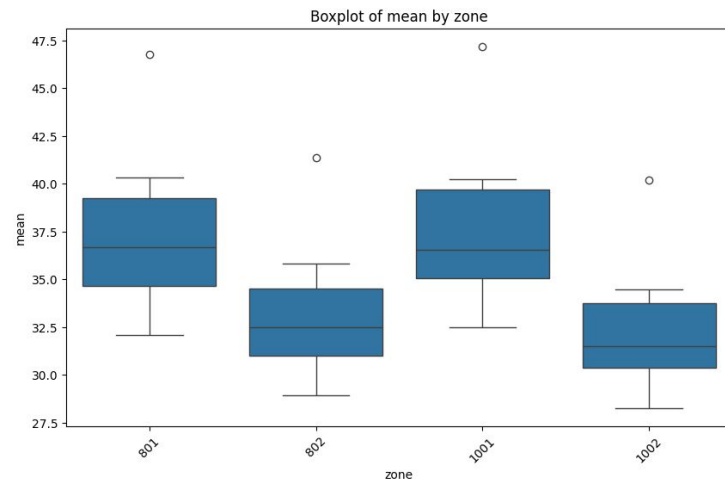
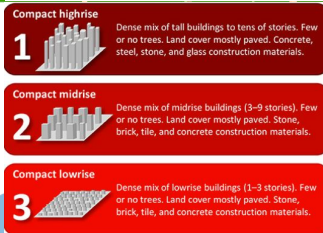
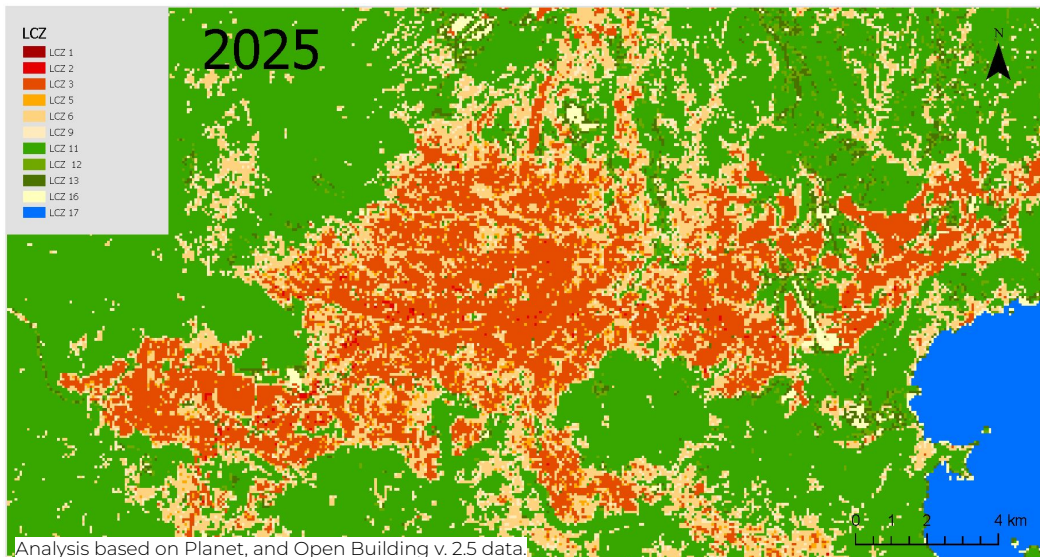


LCZ 2025
LCZ 1
LCZ 2
LCZ 3
LCZ 5
LCZ 6
LCZ 7
LCZ 8
LCZ 9
LCZ A
LCZ B
LCZ C
LCZ D
LCZ E
LCZ F
LCZ G

Analysis based on Planet, and Open Building v. 2.5 data.

3. Results

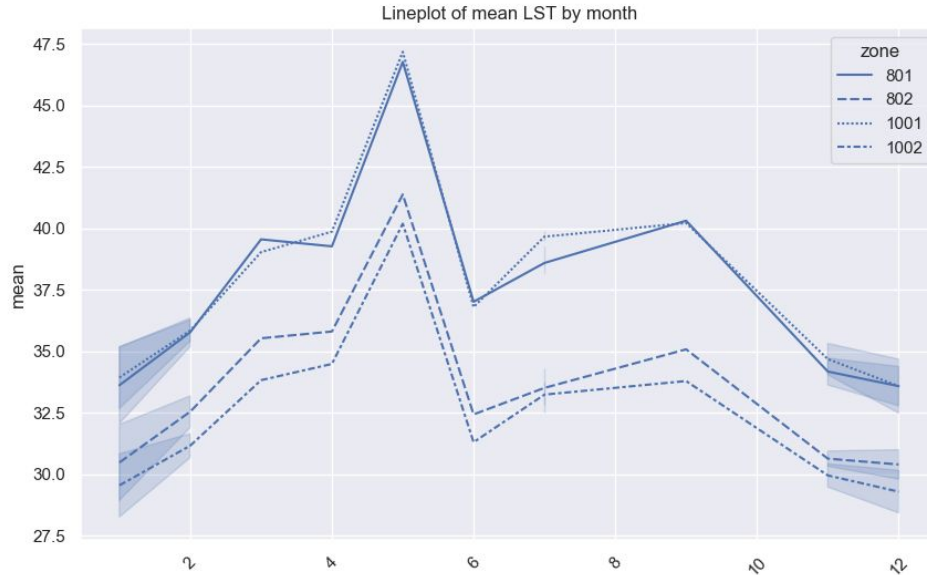
Zonal Statistics and statistical significance between rural and urban areas



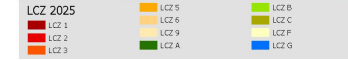
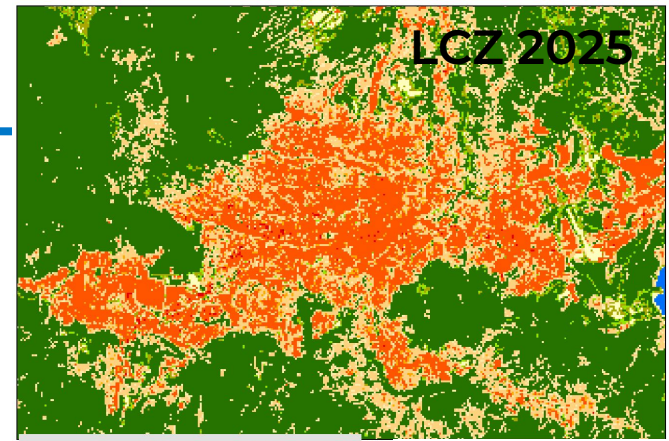
The **results** of this analysis indicated that there is a statistical significance (mean: p-value 0.000, max: p-value 0.0333, minimum: p-value 0.0146) between urban areas (LCZs 1,2,3) and rural areas (LCZs 6, 9).

3. Results

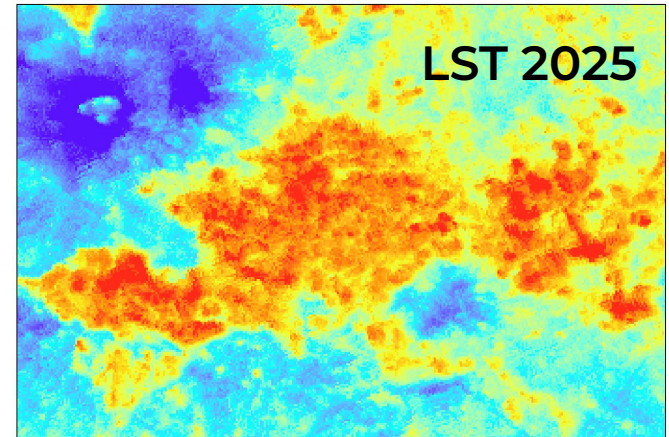
Time-series analysis: LST (mean) by Local Climate Zone for 2025



The **results** show that urban areas (LCZs 1, 2, 3) were ~5 degrees warmer than rural areas (LCZs 6, 9) in 2025. May had the highest LST, in contrast to December with the lowest LST for all LCZs.



Analysis based on Planet, and Open Building 2.5 data.



Analysis based on LST data.

4. Outputs

- The local government of San Salvador will have information on the influence of the different LCZs on land surface temperature and can thereby prioritize mitigation actions in specific LCZs.
- Statistical significance analysis of the comparison of LST between urban areas (LCZ 1,2,3) and rural areas (LCZ 6 and 9) for 2025 for San Salvador.
- Plotting of the results using *boxplot* and *lineplot*, to visualize the distribution of the data and perform a the time series analysis.
- This Python-based code can be used to replicate and upscale the analysis to other years and locations.

